

Scalability & Future Plan

Date	30 June 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	1 Mark

This document outlines how the current **Smart Lender** solution can be scaled to support a larger number of users, increased loan applications, and advanced AI capabilities. It also presents the team's future roadmap for improving the system with additional features and enhanced deployment strategies

Current System Limitations

#	Limitation	Impact	Priority
1	MySQL database hosted on a local server	May experience slower performance with a large number of concurrent users.	Medium
2	Machine Learning model is trained offline	Prediction accuracy depends on periodic retraining with updated datasets.	Medium
3	Manual document verification	Supporting documents are not automatically verified, increasing processing time.	High

Scalability Plan

#	Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a limited number of users on a local server	Deploy the application on cloud platforms such as AWS, Azure, or Render with load balancing.
2	Data Storage	Local MySQL database	Migrate to cloud-managed MySQL or PostgreSQL with automated backup and replication.
3	Performance	Synchronous request processing	Use asynchronous task queues (Celery + Redis) to process prediction requests efficiently.
4	Security	Basic username/password authentication	Implement HTTPS, JWT authentication, rolebased access control, and multi-factor authentication (MFA).

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Deploy Smart Lender on cloud using Docker containers and CI/CD pipeline	Sep 2026	Enables secure online access and supports more concurrent users
Phase 3	Integrate real-time Credit Bureau APIs (e.g., CIBIL) and document verification services	Nov 2026	Improves prediction accuracy and reduces manual verification efforts
Phase 4	Implement automated MLOps pipeline for continuous model retraining using new loan datasets	Feb 2027	Keeps the prediction model updated and improves longterm accuracy.
Phase 5	Develop Android/iOS mobile applications with AI-powered financial recommendations and multilingual support	May 2027	Enhances accessibility, customer experience, and financial guidance for users.